A Note on Triangular Numbers

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Definition 1. The *triangular numbers* $T(n)$ are defined by $T(0) = 0$ and $T(n+1) = T(n) + (n+1)$, so $T(n)$ counts the dots in a triangle with n rows.

Example 1. $T(4) = 10$, since $1 + 2 + 3 + 4 = 10$.

Theorem 1 (Closed form)**.** *For every natural number n, $2\,T(n) = n(n+1)$.*

Proof. By induction on n. The case $n = 0$ is immediate, and $2\,T(n+1) = 2\,T(n) + 2(n+1) = n(n+1) + 2(n+1) = (n+1)(n+2)$. □

Theorem 2. *Let n be a natural number with $n \geq 1$. Then $T(n) + T(n-1) = n^2$.*

Proof. Write $n = k + 1$. Then $T(n) + T(n-1) = 2\,T(k) + (k+1) = k(k+1) + (k+1) = (k+1)^2$ by Theorem 1. □

Example 2. The hypothesis of Theorem 2 is satisfiable: some natural number satisfies $n \geq 1$.